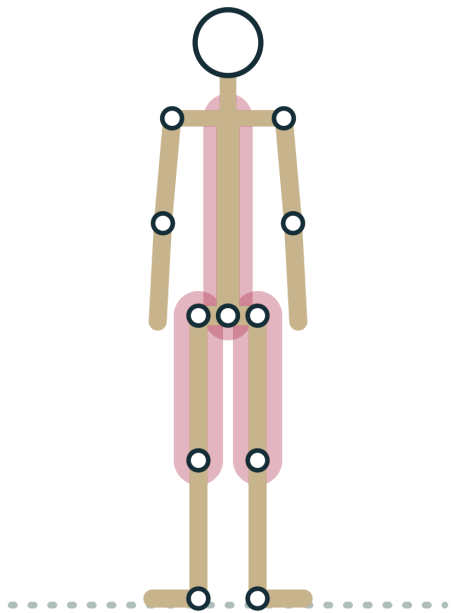


Module 4

Muscles and How They Contract

Knowing how muscles shorten, lengthen, share roles and respond to stretch lets you cue strength and flexibility honestly and safely.



3 hours • 7 lessons • Dr. Akash Garg, D.C.
Anatomy & Physiology for Vinyasa Teachers

Overview

Every pose you teach is built from muscles pulling on bones. This module starts inside the muscle, with the sliding filaments of the sarcomere, and works outward to the three contraction types you meet in every class: the eccentric lower into Chaturanga, the isometric hold of Chair and the concentric press back up. You will learn to name the roles muscles play in a movement, agonist, antagonist, synergist and stabilizer, and to work them out for any pose by asking which joint moves and where gravity pulls.

The second half turns to stretching, where yoga lore and science often part ways. You will learn what muscle spindles and Golgi tendon organs actually do, why the classic explanation for PNF stretching is now questioned, and why stretch tolerance is the mainstream explanation for short-term flexibility gains. You will also meet fascia as a real and interesting tissue, without the unsupported claims. The module ends with a pose lab that maps muscle roles in Warrior III and Bridge.

By the end of this module you can

1. Describe the structure of skeletal muscle from whole muscle to sarcomere and explain the sliding filament model in plain words
2. Identify concentric, eccentric and isometric contractions in Chaturanga, Chair and other common poses
3. Assign agonist, antagonist, synergist and stabilizer roles for a given joint action in a pose
4. Explain the stretch reflex, reciprocal inhibition and the Golgi tendon organ, and state what current evidence says about how PNF works
5. Distinguish the nervous system and tissue contributions to flexibility and cue end-range strength for flexible students
6. Separate supported from unsupported claims about fascia and choose accurate cueing language

Lessons

- **4.1** Muscle structure and how a muscle shortens (25 min)
- **4.2** Concentric, eccentric and isometric: lowering through Chaturanga, holding Chair (30 min)
- **4.3** Agonist, antagonist, synergist and stabilizer (25 min)
- **4.4** Reciprocal inhibition, the stretch reflex and the Golgi tendon organ (35 min)
- **4.5** What flexibility is: nervous system tolerance and tissue length (25 min)
- **4.6** Fascia without the hype (20 min)
- **4.7** Pose lab: muscle roles in Warrior III and Bridge (20 min)

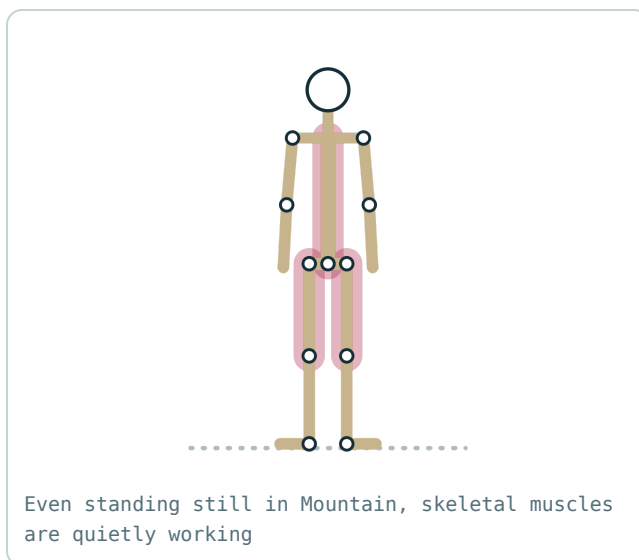
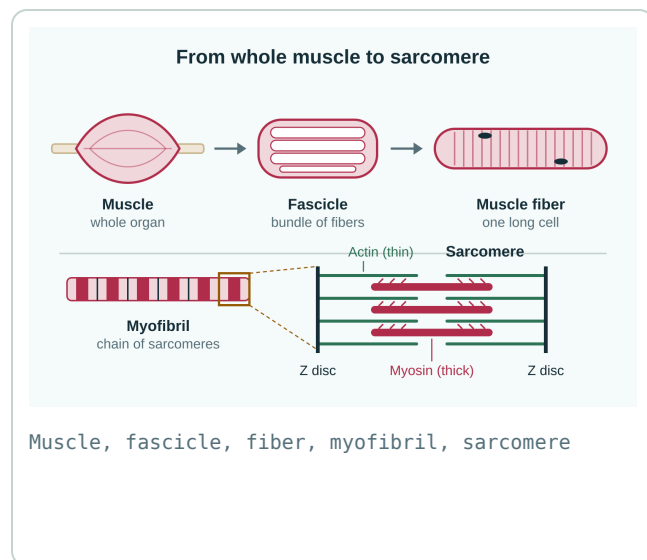
4.1 Muscle structure and how a muscle shortens

Skeletal muscle is the voluntary, striated muscle that attaches to bone through tendons. It is the only one of the three muscle types (skeletal, cardiac and smooth) that a yoga teacher can cue directly. Its jobs are to move joints, to hold positions against gravity and to produce heat. Even a still pose like Mountain involves constant low-level muscle work.

Muscle is organized like boxes inside boxes. The whole muscle is made of bundles called fascicles. Each fascicle contains many muscle fibers, and each fiber is a single long cell with many nuclei. Inside each fiber are myofibrils, and each myofibril is a chain of sarcomeres. The sarcomere, running from one Z disc to the next, is the basic unit that shortens.

The sliding filament model explains how. Thin actin filaments are anchored to the Z discs, and thick myosin filaments sit between them. When a nerve signal arrives, calcium is released inside the fiber, which lets myosin heads attach to actin. The heads pivot and pull actin toward the center of the sarcomere, then use ATP to detach and reattach further along. The filaments do not shorten; they slide past each other, so the Z discs are drawn together and the sarcomere, fiber and whole muscle shorten.

Force is graded through motor units. A motor unit is one motor neuron and all the fibers it supplies; when it fires, all those fibers contract. The nervous system produces more force by recruiting more motor units and firing them faster, and it starts with small units, which allows fine control. Finally, muscles can only pull. Because no muscle can push, every joint needs muscles on opposite sides, or gravity, to move it back the other way.



KEY POINTS

- Skeletal muscle is voluntary, striated and attached to bone by tendons
- Hierarchy: muscle, fascicle, fiber, myofibril, sarcomere
- Myosin pulls actin toward the sarcomere center; the filaments slide, they do not shrink
- Calcium switches contraction on; ATP lets myosin release and reset
- Force is graded by recruiting more motor units and firing them faster
- Muscles only pull, so movement back the other way needs another muscle or gravity

FEEL IT

Stand in Mountain with a hand resting on the front of each thigh. Bend the knees a little into a small Chair and feel the quadriceps firm under your hands. Sink a little deeper and notice them firm further as more motor units are recruited. Stand up again and feel them soften. Repeat slowly three times, breathing steadily.

CUE IT

- “Bend the knees just a little and feel the front of the thighs wake up”
- “Let the effort build gradually, like turning up a dimmer switch, not flicking a light on”
- “Stand tall and notice how little muscle it takes when the bones are stacked”

CHECK IT

During a contraction, what happens inside the sarcomere?

Answer: Myosin pulls actin toward the center so the filaments slide past each other. In the sliding filament model the filaments keep their length and slide past each other, drawing the Z discs closer together.

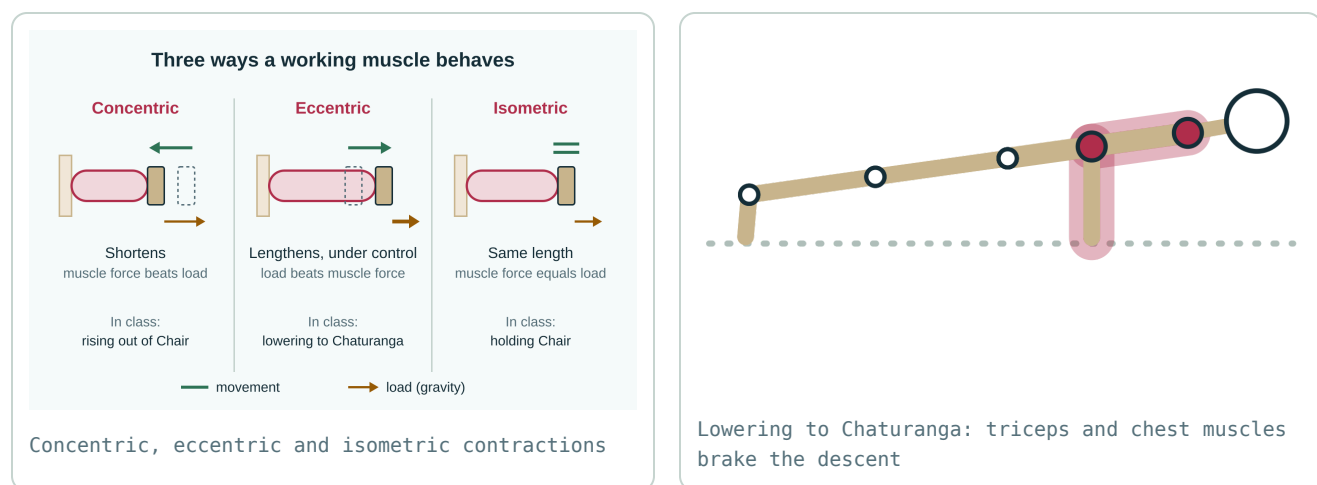
4.2 Concentric, eccentric and isometric: lowering through Chaturanga, holding Chair

In anatomy, a contraction means a muscle is actively producing tension. It does not always mean the muscle gets shorter. There are three types, and you can identify them by comparing the muscle's force with the load, which in yoga is almost always gravity acting on body weight.

In a concentric contraction the muscle shortens while it works, because its force is greater than the load. Rising out of Chair and pressing from Chaturanga back up to plank are concentric for the quadriceps and triceps respectively. In an eccentric contraction the muscle lengthens while it works, because the load is greater than its force and the muscle is acting as a brake. Lowering from plank into Chaturanga is eccentric for the triceps, pectoralis major and front deltoid. Lowering into Chair is eccentric for the quadriceps and gluteus maximus. In an isometric contraction the muscle works without changing length, because force and load are matched. Holding Chair or pausing halfway down in Chaturanga are isometric.

A reliable rule: when a body part moves against gravity, the muscles producing that motion work concentrically. When it moves with gravity at a controlled speed, the muscles that would move it the other way work eccentrically. This means the same muscle usually does both the lifting and the lowering. The triceps both press you up and lower you down in Chaturanga.

Eccentric contractions produce more force for less energy than concentric ones, which is why most people can lower more slowly than they can press up. Unfamiliar eccentric work is also the main cause of delayed onset muscle soreness, which peaks one to two days later. Isometric holds build strength at a specific joint angle and are easy to scale by depth, but students often hold their breath in them, so cue steady breathing.



KEY POINTS

- Contraction means producing tension, not necessarily shortening
- Concentric: shortens, force beats load. Eccentric: lengthens, load beats force. Isometric: same length
- Lowering to Chaturanga is eccentric for the triceps; holding Chair is isometric for the quadriceps
- Ask where gravity pulls to identify the contraction type
- Eccentric work is strongest and causes the most delayed soreness when unfamiliar
- Cue breathing during strong isometric holds

FEEL IT

Come to plank with knees down or lifted. Shift forward and lower for a slow count of four, feeling the triceps and chest work as the elbows bend. Pause halfway for three breaths, then press back up to plank. Notice which phase felt strongest and which felt hardest.

CUE IT

- “Lower slowly for a count of four, letting the backs of the arms control the descent”
- “Stay here in Chair and keep breathing; the thighs are working even though nothing moves”
- “Press the floor away to rise, and feel the same muscles now lifting you”

CHECK IT

A student lowers slowly from plank into Chaturanga. What are the triceps doing?

Answer: Contracting eccentrically, lengthening under control. Gravity is bending the elbows and the triceps lengthen while working to control the descent, which is an eccentric contraction.

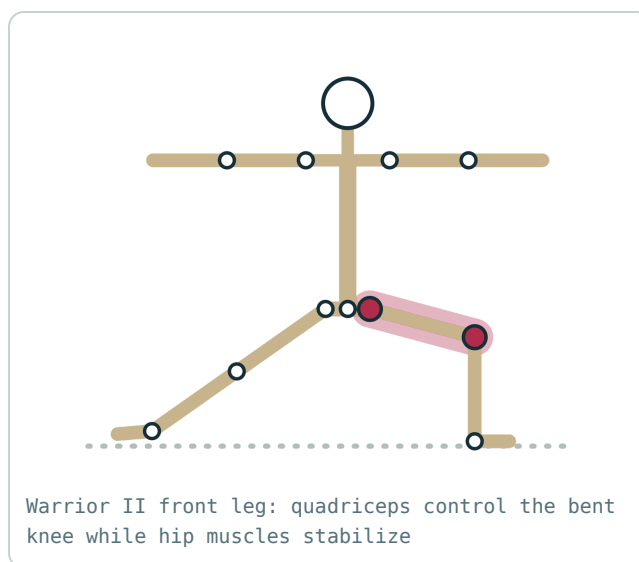
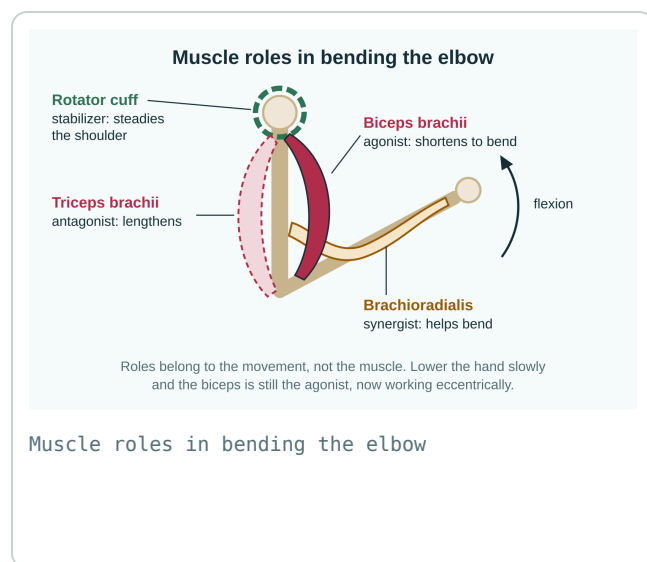
4.3 Agonist, antagonist, synergist and stabilizer

Because muscles can only pull, they work in teams, and anatomists describe four roles within any movement. The agonist, or prime mover, is the main muscle producing the movement. The antagonist produces the opposite movement and usually lengthens to allow the agonist's action. A synergist assists the agonist, either by adding force or by cancelling an unwanted secondary action. A stabilizer, also called a fixator, holds a nearby joint steady so the agonist has a firm base to pull from.

At the elbow, the biceps brachii and brachialis are agonists for bending, the triceps is the antagonist, the brachioradialis is a synergist and the rotator cuff and shoulder blade muscles stabilize the shoulder. In Chair, the quadriceps are the agonist at the knee in every phase: lowering eccentrically, holding isometrically and rising concentrically. The hamstrings are antagonists at the knee, while the gluteus medius and deep hip rotators stabilize the thigh so the knee tracks over the second toe.

These roles belong to the movement, not the muscle. The biceps is the agonist for bending the elbow and the antagonist when straightening it against resistance. Muscles that cross two joints, like the hamstrings and rectus femoris, can play different roles at each joint at the same time. The dependable method is to name the joint, name the movement and the direction of gravity, and then ask which muscles produce and which oppose it.

Stabilizers often decide whether a pose looks effortless. Serratus anterior keeps the shoulder blades on the ribs in Down Dog, the rotator cuff centers the head of the humerus, and deep abdominals stiffen the trunk before the limbs move. When you see wobble or collapse, the problem is often a stabilizer. Use anatomy to choose your cue, but deliver it as a plain action.



KEY POINTS

- Agonist produces the movement; antagonist opposes it and usually lengthens
- Synergists assist; stabilizers hold nearby joints steady
- In Chair the quadriceps are the knee agonist in all three phases
- Roles depend on the movement: name the joint and action first
- Two-joint muscles can play different roles at each joint
- Cue the action a stabilizer produces rather than naming the muscle

FEEL IT

Stand on one leg near a wall and straighten the standing knee firmly. Feel the quadriceps harden with one hand and the softer hamstrings with the other. Without moving, press the standing heel down and slightly back as if to bend the knee, and feel the hamstrings switch on. Change sides.

CUE IT

“Draw the front knee toward the little-toe side of the foot”

“Press the floor away so the shoulder blades wrap around the ribs”

“Keep the standing thigh firm and let the back of the leg stay soft”

CHECK IT

In Chair, which muscle group is best described as the agonist at the knee?

Answer: Quadriceps. The quadriceps control the knee against gravity in every phase of Chair, so they are the agonist; the hamstrings are the antagonist at the knee.

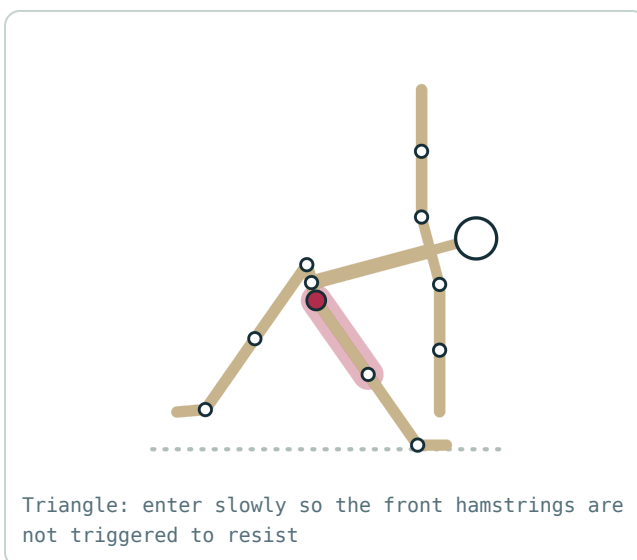
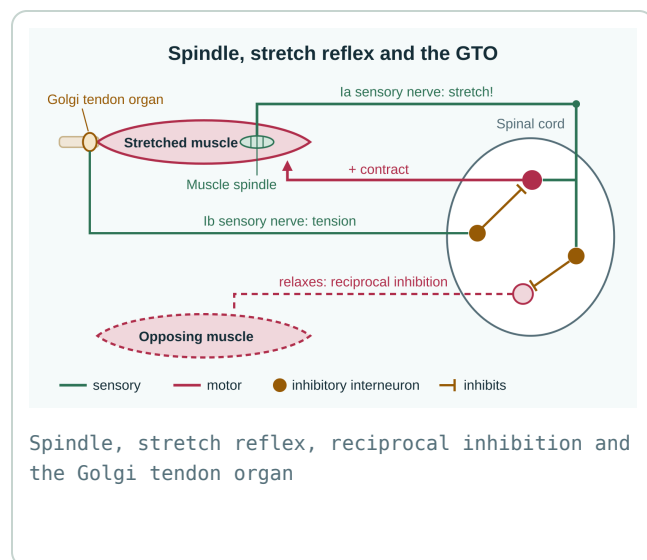
4.4 Reciprocal inhibition, the stretch reflex and the Golgi tendon organ

Muscles contain two sensors that shape stretching. Muscle spindles lie parallel to the fibers in the muscle belly and sense length and the speed of stretch. Golgi tendon organs (GTOs) lie in series at the muscle-tendon junction and sense tension, particularly tension produced by contraction. Both send information to the spinal cord and contribute to proprioception, your sense of where your body is.

The stretch reflex begins in the spindle. A fast stretch fires its Ia sensory nerve, which connects directly to the motor neurons of the same muscle and makes it contract. This one-synapse loop is the knee-jerk reflex. It is highly sensitive to speed, which is why slow entry into stretches keeps it quiet and why bouncing is discouraged in general classes. The same Ia signal also excites an inhibitory interneuron that damps the motor neurons of the opposing muscle. This is reciprocal inhibition. It is well established and allows opposing muscles to move smoothly without fighting. Yoga teachers use it as a cue, such as engaging the quadriceps to help the hamstrings release, but in a slow relaxed stretch the target muscle is usually quiet already, so the extra effect is probably modest.

The GTO's Ib nerve works through an inhibitory interneuron to reduce activity in the same muscle, called autogenic inhibition. It was once described as an emergency cutoff, but GTOs respond to small forces and give continuous feedback during normal movement.

PNF contract-relax stretching, where you contract the stretched muscle, relax, then stretch further, reliably improves short-term range. It has long been explained by GTO autogenic inhibition, but current research questions this: post-contraction inhibition is very brief and the muscle does not stay measurably more relaxed. The leading explanation is increased stretch tolerance. Teach PNF if you find it useful, but present the mechanism honestly.



KEY POINTS

- Spindles sense length and speed; GTOs sense tension
- Stretch reflex: fast stretch, spindle fires, same muscle contracts
- Reciprocal inhibition damps the antagonist; real, but its added benefit in slow stretches is modest
- GTOs give continuous force feedback, not just an emergency cutoff
- PNF improves short-term range; the GTO explanation is questioned and stretch tolerance is the leading explanation
- Enter stretches slowly and use moderate effort for contract-relax

FEEL IT

Sit in a forward fold with knees bent as much as you need. Press the heels into the floor at about a third of your strength for five seconds, then relax completely on a slow exhale. Fold a little further, noticing whether the stretch feels less intense at the same depth. Repeat no more than three times.

CUE IT

“Move into the stretch slowly and stop at the first clear sensation”

“Press the heels gently into the floor, then let everything go on the exhale”

“Stay where you can breathe smoothly; no bouncing”

CHECK IT

Which statement about PNF contract-relax stretching best fits current evidence?

Answer: It improves short-term range mainly by increasing tolerance to the stretch. PNF reliably improves short-term range, but the GTO inhibition explanation is questioned; increased stretch tolerance is the leading explanation.

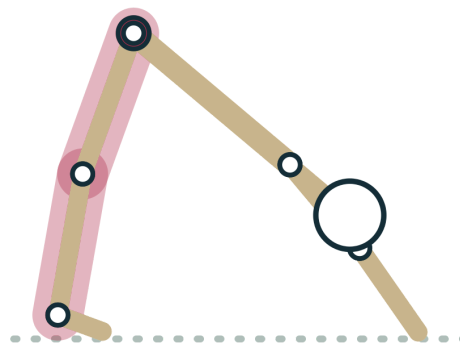
4.5 What flexibility is: nervous system tolerance and tissue length

Flexibility, the range of motion available at a joint, has two ingredients. One is the tissue: the muscle-tendon unit, the joint capsule, ligaments, skin and nerves, and how stiff they are. The other is the nervous system: how much stretch sensation a person tolerates before stopping. In most people the end of a stretch is set by sensation, not by tissue reaching its limit. Range often increases under general anesthesia, which shows how much the nervous system contributes.

Within a single session, range increases quickly, but the force needed to reach a given angle barely changes, meaning the muscle has not become meaningfully longer or less stiff. What changes is stretch tolerance. This sensory theory, or stretch tolerance model, is the current mainstream explanation for short-term flexibility gains. These gains fade within hours. Over weeks of regular stretching, range increases more durably. Tolerance remains a large part of that, and some studies also report tissue changes such as reduced stiffness or longer fascicles, though the evidence is mixed. Consistency matters more than intensity, and range tends to slip back when practice stops.

Not every limit is a tight muscle. Bone shape varies widely, especially at the hip, and sets hard limits no stretch can change. Compression, where tissue meets tissue as in Child's pose, is another limit, and the joint capsule, ligaments and nerves can also restrict range. A stretch is felt along the length of a tissue; a hard stop with pinching at the front of a joint suggests compression and should not be pushed.

For very flexible or hypermobile students, the goal is control rather than more range. Teachers do not diagnose hypermobility but can cue students to hold back from end range and make the pose strong. For stiff students, active work at the end of range builds more usable flexibility than passive hanging.



Down Dog range reflects both tissue and tolerance

KEY POINTS

- Flexibility depends on tissue properties and on nervous system tolerance
- Short-term gains come mainly from increased stretch tolerance (sensory theory)
- Long-term gains need regular practice; tissue changes are real but modest and debated
- Bone shape and compression set limits that stretching cannot change
- A pinching hard stop is not a stretch
- Hypermobile students need control and strength at end range, not more range

FEEL IT

In a seated forward fold, stop at the first clear stretch and note where your hands reach. Rate the intensity from 0 to 10, breathe slowly for a minute, then rate it again at the same depth. Move to a new depth that matches your first rating and note the change. Reflect on what actually changed in that minute.

CUE IT

“Stop at the first clear stretch and let the breath slow down”

“Make the pose strong before you make it deep”

“If you feel a pinch at the front of the hip, come out a little and change the angle”

CHECK IT

A student reaches further in a forward fold after holding it for a minute. What is the most likely main reason?

Answer: The nervous system now tolerates more stretch sensation. Short-term range gains are best explained by increased stretch tolerance; muscle length and stiffness change little in one session.

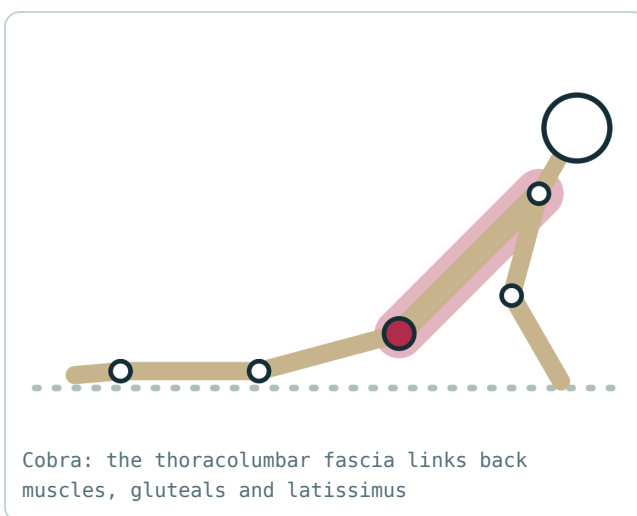
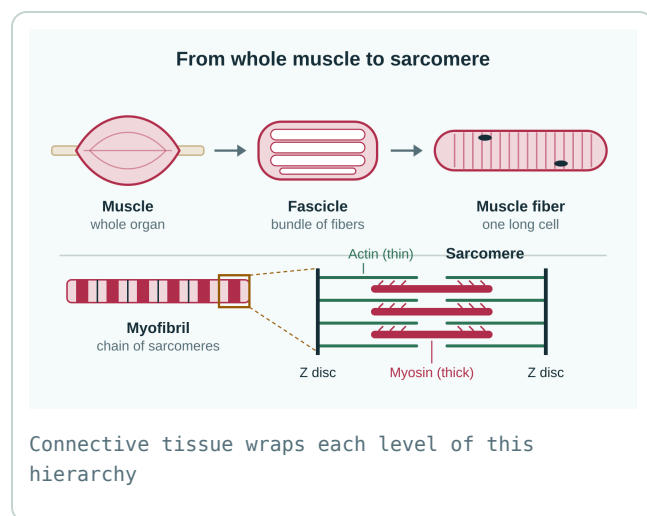
4.6 Fascia without the hype

Fascia is connective tissue, made mostly of collagen fibers in a watery ground substance, that wraps, separates and links structures throughout the body. Every level of muscle is wrapped: the whole muscle by the epimysium, each fascicle by the perimysium and each fiber by the endomysium, and these layers merge to form tendons. Deep fascia forms tough sheets such as the thoracolumbar fascia in the low back, the fascia lata and iliotibial band on the thigh, and the plantar fascia under the foot. Superficial fascia is a looser layer under the skin mixed with fat.

Fascia has real, well supported functions. It transmits force from muscle to bone and sideways between neighboring muscles. It separates structures and lets them glide, as when skin slides over the thigh as you bend the knee. It is richly supplied with sensory nerve endings, so it contributes to proprioception and can be a source of pain. Like tendon and ligament, it adapts slowly to regular loading over months.

Several popular claims are not supported. Studies estimating the force needed to deform dense fascia such as the iliotibial band found it far greater than hands or a foam roller can apply, so the idea of releasing, melting or permanently lengthening fascia is not supported. The real short-term sense of release after rolling or massage is better explained by nervous system changes and fluid shifts. The claim that fascia stores emotions or trauma has no scientific support, and neither does the idea that particular poses rehydrate fascia. Some anatomical continuity along proposed myofascial lines has been demonstrated, but its importance for movement is debated.

For teachers, the practical lesson is language. Describe sensation, which is accurate, rather than promising changes to tissue. Say the back of the body feels long, not that the fascia has released. When students feel emotional in a pose, respect the experience without explaining it with an unsupported tissue claim.



KEY POINTS

- Fascia is collagen-rich connective tissue that wraps every level of muscle
- It transmits force, lets structures glide and is richly innervated
- Hands and rollers cannot release or permanently lengthen dense fascia
- Short-term release after rolling or massage is mostly a nervous system effect
- No evidence that fascia stores emotions or that poses rehydrate it
- Describe sensation in cues rather than promising tissue changes

FEEL IT

Sitting, pinch a fold of skin on the front of your thigh and slowly bend and straighten the knee, feeling the layers glide. Then stand in Mountain and reach one arm long overhead, noticing how sensation spreads along the side body. Describe what you feel in your own words without using the word release.

CUE IT

“Notice the sensation spread along the whole back of the body”

“Let the side of the body feel long as you reach”

“Rolling may help you feel looser; notice how you move afterwards”

CHECK IT

Which statement about fascia is best supported by current evidence?

Answer: Fascia transmits force and contains many sensory nerve endings. Force transmission and rich innervation are well supported; permanent deformation by rolling and emotional storage are not.

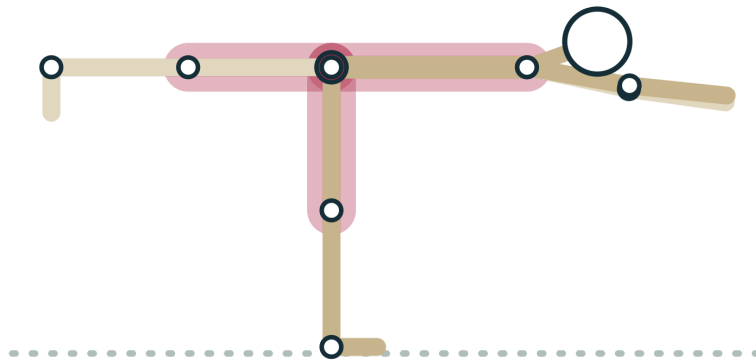
4.7 Pose lab: muscle roles in Warrior III and Bridge

Warrior III is largely a whole-body isometric pose. On the standing leg, the quadriceps hold the knee straight or slightly soft and the muscles of the lower leg and foot make constant balance corrections. The pelvis tips forward over the standing thigh, which is hip flexion, and gravity would pull it further, so the standing gluteus maximus and hamstrings control that tilt: eccentrically on the way in, isometrically in the hold. The standing gluteus medius is the key stabilizer, keeping the pelvis level; when it does not hold, the lifted hip rolls up toward the ceiling. On the lifted side, the gluteus maximus and hamstrings are agonists holding hip extension against gravity, while the hip flexors are antagonists and lengthened. The erector spinae hold the trunk level, the deep abdominals stiffen it and the front of the shoulders holds the arms forward.

Useful cues target these roles in plain words: press down through the standing foot, reach back through the lifted heel and keep both hip points facing the floor. Hands on blocks or a wall reduce the load on the back and make balance easier.

Bridge is a hip extension pose. Lifting the pelvis is concentric work for the gluteus maximus and hamstrings, holding is isometric and lowering slowly is eccentric for the same muscles. The quadriceps help push the knees forward, the adductors stabilize the knees from splaying and the erector spinae support the arch. The hip flexors are the antagonists, lengthened across the front of the hips. Hamstring cramping is common because the hamstrings are working at a short length, with the knee bent and hip extended; lower down and reset rather than pushing on. In Bridge the neck is flexed under load, so keep the head still.

Working through a pose this way, joint by joint and phase by phase, is the method you can apply to any pose you teach.



Warrior III: standing leg vertical, trunk and back leg level

KEY POINTS

- Warrior III is mostly isometric: standing leg, lifted leg and trunk all hold against gravity
- Standing gluteus medius keeps the pelvis level; a rolling hip signals it needs help
- Gluteus maximus and hamstrings are agonists on both sides in Warrior III
- In Bridge the gluteals and hamstrings lift concentrically, hold isometrically and lower eccentrically
- Adductors stabilize the knees in Bridge; hip flexors are lengthened antagonists
- Keep the head still in Bridge and reset if the hamstrings cramp

FEEL IT

Lie on your back with knees bent and feet hip-width apart. Lift the hips over four counts, hold for three breaths with the head still, then lower one segment at a time over four counts. Repeat with a block between the thighs and notice the adductors working. Name the agonist, antagonist and stabilizer you felt.

CUE IT

“Press down through the standing foot and reach back through the lifted heel”

“Keep both hip points facing the floor”

“Press into the feet and lift from the back of the pelvis, keeping the chin still”

CHECK IT

In Warrior III, the lifted-leg hip keeps rolling up toward the ceiling. Which muscle role most needs attention?

Answer: The standing leg's gluteus medius as a stabilizer. The standing gluteus medius stabilizes the pelvis; when it is not holding, the pelvis rotates and the lifted hip opens toward the ceiling.

Key terms

Skeletal muscle	Voluntary, striated muscle attached to bone by tendons; the quadriceps holding you in Chair are skeletal muscle.
Fascicle	A bundle of muscle fibers wrapped in connective tissue; the visible grain in a muscle such as the hamstrings.
Muscle fiber	A single long muscle cell with many nuclei; thousands of them work together when you press up from Chaturanga.
Myofibril	A thread inside a muscle fiber made of sarcomeres linked end to end.
Sarcomere	The basic contractile unit of muscle, from one Z disc to the next; millions shorten together when you rise out of Chair.
Actin and myosin	The thin (actin) and thick (myosin) protein filaments; myosin heads pull actin to shorten the sarcomere.
Motor unit	One motor neuron and all the fibers it controls; small motor units make the fine ankle corrections in Tree.
Concentric contraction	A muscle shortens while working; the quadriceps as you rise out of Chair.
Eccentric contraction	A muscle lengthens while working, acting as a brake; the triceps as you lower into Chaturanga.
Isometric contraction	A muscle works without changing length; the quadriceps while holding Chair.
Agonist (prime mover)	The main muscle producing a movement; the gluteus maximus lifting the pelvis in Bridge.
Antagonist	The muscle producing the opposite movement, usually lengthening; the hip flexors in Bridge.
Synergist	A muscle that assists the agonist or cancels an unwanted action; the brachioradialis helping bend the elbow.
Stabilizer	A muscle that holds a nearby joint steady; the standing gluteus medius keeping the pelvis level in Warrior III.
Muscle spindle	A stretch sensor in the muscle belly that detects length and speed of stretch; it fires if you drop fast into a fold.
Golgi tendon organ	A tension sensor at the muscle-tendon junction; it reports force as you press the heels down in contract-relax stretching.
Stretch reflex	A fast spinal reflex in which a quick stretch makes the same muscle contract; the reason bouncing in Triangle backfires.
Reciprocal inhibition	Damping of the antagonist's motor neurons when a muscle contracts; engaging the quadriceps in Down Dog damps the hamstrings slightly.

PNF stretching	Proprioceptive neuromuscular facilitation: contract the stretched muscle, relax, then stretch further, as in a seated fold with heels pressing down.
Stretch tolerance	How much stretch sensation a person accepts; the main reason you reach further after a minute in a forward fold.
Fascia	Collagen-rich connective tissue that wraps and links muscles; the thoracolumbar fascia spans the low back in Cobra.
Hypermobility	Joints that move beyond typical range; a front knee locking back in Triangle may be a sign, and calls for control cues.

Myths to retire

Muscles shorten because the filaments inside them shrink

Actin and myosin keep their length and slide past each other, drawing the Z discs closer together.

A muscle that is lengthening must be relaxed

Lengthening muscles are often working hard eccentrically, like the triceps lowering you into Chaturanga.

PNF works because the Golgi tendon organ switches the muscle off

PNF improves short-term range, but the GTO explanation is questioned; increased stretch tolerance is the leading explanation.

Holding a stretch makes the muscle longer within that session

Short-term gains come mainly from greater tolerance of the stretch sensation; muscle length and stiffness change little.

Yoga or foam rolling releases fascia, and fascia stores emotions

Dense fascia cannot be deformed by hands or rollers, the felt release is mostly neural, and there is no evidence fascia stores emotions.

Self check

1. Explain the sliding filament model in two sentences.
2. Describe the three contraction types in Chair.
3. Why does the triceps work while the elbow bends in Chaturanga?
4. Name the agonist, antagonist and a stabilizer for hip extension in Bridge.
5. Compare the muscle spindle and the Golgi tendon organ.
6. What does current evidence say about why PNF improves range?
7. What is the sensory theory of flexibility?
8. Give one supported and one unsupported claim about fascia.

Model answers

1. Myosin heads attach to actin, pivot and pull the thin filaments toward the center of the sarcomere, using calcium to switch on and ATP to release. The filaments slide past each other without shortening, so the Z discs come closer and the muscle shortens.
2. Lowering into Chair is eccentric for the quadriceps and gluteals, holding is isometric, and rising is concentric. The same muscles work in all three phases.
3. Gravity is bending the elbow, so the triceps must lengthen under tension to control the descent. That is an eccentric contraction acting as a brake.
4. The gluteus maximus (with the hamstrings) is the agonist, the hip flexors are the antagonists and the adductors stabilize the knees.
5. The spindle lies in the muscle belly and senses length and speed of stretch, exciting the same muscle. The GTO lies at the muscle-tendon junction, senses tension and inhibits the same muscle.
6. PNF reliably improves short-term range, but the idea that GTO inhibition relaxes the muscle is questioned. Increased stretch tolerance is the leading explanation.
7. It proposes that short-term range gains come from greater tolerance of stretch sensation rather than longer or less stiff muscles. It is the mainstream explanation today.
8. Supported: fascia transmits force and is richly innervated. Unsupported: hands, rollers or poses can release dense fascia or that fascia stores emotions.